

Contour Integration via Residue Theorem and Cauchy's Integral Formula

Source: J. E. Santos, Cambridge Part IB Complex Methods, Problem Sheet 2, Question 9.

1. By integrating the function

$$z^n(z-a)^{-1}(z-a^{-1})^{-1}$$

around the unit circle and applying the residue theorem, evaluate

$$\int_0^{2\pi} \frac{\cos n\theta}{1-2a\cos\theta+a^2} d\theta, \quad (1.1)$$

where a is real, $a > 1$, and n is a non-negative integer.

2. Obtain the same result using Cauchy's integral formula instead of the residue theorem.

Solution. Parametrise the integration by $z = e^{i\theta}$ with $\theta \in [0, 2\pi)$. The integral becomes

$$I = \int_0^{2\pi} d\theta \frac{ae^{in\theta}}{-1+2a\cos\theta-a^2} = \int_0^{2\pi} d\theta \frac{ae^{-in\theta}}{-1+2a\cos\theta-a^2} = -2\pi i \frac{a^{-n+1}}{a^2-1} \quad (1.2)$$

Therefore,

$$\int_0^{2\pi} \frac{\cos n\theta}{1-2a\cos\theta+a^2} d\theta = -\frac{I}{ai} = 2\pi \frac{a^{-n}}{a^2-1}. \quad (1.3)$$